Question 1

The three-bus bar electricity transmission system presented in the Figure 1.1, consists of two generation units, connected to busbars 1 and 3, which have a rating and a marginal cost (MC) as depicted in the figure. These two generators supply the loads located at buses 2 and 3.

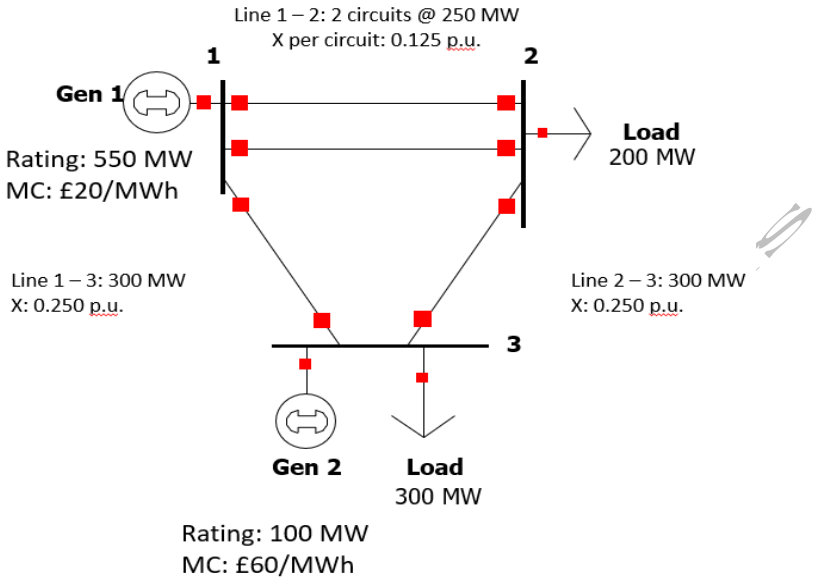


Figure 1.1 Three-bus bar electricity transmission system

Generation, demand and transmission circuit data are presented in corresponding tables below:

	Rating (MW)	Marginal Cost (£/MWh)
Gen 1	550	20
Gen 2	100	60

	Rating (MW)
Load (bus 2)	200
Load (bus 3)	300

	Rating per circuit (MW)	Reactance per circuit (pu)
Line 1-2	250	0.125
Line 1-3	300	0.250
Line 2-3	300	0.250

a) Determine the economic dispatch ignoring transmission line constraints. Calculate the corresponding cost and power flows.

[3]

Output of each generator should be determined while minimising the total generation cost.

$\min \sum_{i=1}^{G} c_i Pg_i$

$\sum Pg_i = \sum Pd$

$Pg_i \leq Pg_i^{max} \quad \forall i \in G.$

Solution: PG1 = 500MW, PG2= 0 MW and cost = £ 10,000/h.

[1]

The flows can be determined by applying superposition concept: we consider only the load at bus 2 which is 200 MW. Hence, Gen1 output is 200 MW. The power flow through each of the two circuits of line 1-2 is equal to ½ *200*(0.25+0.25)/ (0.25+ 0.25 + 0.125*0.125/(0.125+0.125)) = 88.9 MW. Thus, the flow 1-3-2 is equal to 200 – 2*88.9 = 22.2MW.

Then, we consider only the load at bus 3, i.e. 300 MW. In this case, GEN1 outputs 300 MW. In this case, the flow through 1-3 is 00 (0.25 + 0.125*0.125/(0.125+0.125)

) / (0.25 + 0.25 +0.125*0.125/(0.125+0.125)) = 166.7 MW. Thus from each circuit of 1-2 we have (300- 166.7) /2 =66.7MW.
We then add the corresponding flows and we get:
Line flow 1-3 is: 166.7 + 22.2 = 188.9 MW
Line flow 1-2 (over each circuit): 88.9 + 66.7 = 155.6 MW
Line flow 2-3: 133.3- 22.2 = 111.1 MW

[2]

b) Taking into account the capacity of transmission circuits, find the power flows when there is an outage at one of the lines as follows:

i) When line 1-3 is out of operation
ii) When line 2-3 is out of operation
iii) When one of the two circuits of line 1-2 is out of operation

[4]

Line 1-3 out of service: In this case the flow over 1-3 is equal to 0 MW.
Also we noted that PG1 = 500 MW and PG2 = 0MW. Hence the 500MW pass from line 1-2, i.e. 250 MW from each of the two circuits (since they have the same reactance, the flows per circuit are equal in magnitude). Hence from line 2-3 , the flow is equal to 500 – 200 = 300 MW. Thus, we reach the maximum limit of lines 1-2 and 2-3. The lines are not overloaded.

[1]

Line 2-3 out of service: In this case the flow over 2-3 is equal to 0 MW.
Also we noted that PG1 = 500 MW and PG2 = 0MW. Hence 300MW are sent to feed the load at bus 3, and 200 MW are sent over 1-2 (i.e. 100 MW per circuit) to feed the load at bus 2. Thus, we reach the maximum limit of line 2-3. The lines are not overloaded.

[1]

Line 1-2 out of service (only one circuit): We noted that PG1 = 500 MW and PG2 = 0MW. In this case, we can find the flows by considering load at bus 2 and then load at bus 3 and adding the corresponding flows.
-Considering only load at bus 2: line 1-2 flow:
200*(0.25+0.25)/(0.25+0.25+0.125)=160 MW and thus flow over 1-3-2 is 200-160 = 40MW.
-Considering only load at bus 3: flow 1-2-3 is: 300*(0.25)/(0.125 + 0.25+0.25) = 120 MW and thus flow 1-3 is : 300 – 120 = 180 MW.
Thus in total: flow 1-2 is: 120 + 160 = 280 MW. Flow 2-3: 120 – 40 = 80 MW.
Flow 1-3: 180 + 40= 220 MW. Since each circuit of line 1-2 is 250 MW, and we send 280 MW, the lines are overloaded.

[2]

c) For the case of question b), if you find that one of the lines is overloaded, re-dispatch generation to make sure that none of the lines would be overloaded. Calculate corresponding generation operation costs and explain the differences against a).

[5]

As line 1-2 would be overloaded, we need to re-dispatch generation (to ensure that none of the lines would be overloaded).
Sensitivity of flows with respect to the changes in power injection can be quantified. In the first case, the PG1 = 200MW feeding the load at bus 2 , and the flow over 1-2 is 160 MW. In the second case, PG1=300 MW feeding the load at bus 3, and the flow over 1-2 is 120 MW. Hence, the sensitivities are:

$$\frac{\Delta F_{12}}{\Delta P_2} = \left(-\frac{160}{200}\right) = -0.8, \quad \frac{\Delta F_{12}}{\Delta P_3} = \left(-\frac{120}{300}\right) = -0.4$$

Criteria for determining the output of generator 2 that is needed to alleviate the overloading of line 1-2: minimize Pg_2 subject to the constraint $F_{12}^{(0)} + \frac{\Delta F_{12}}{\Delta P_3} Pg_2 \leq 250$ $i.e.$ $280 - 0.4Pg_2 \leq 250$ → PG2= 75MW i.e the out-of merit generator Gen2 needs to generate 75MW.

In this case PG1= 425 MW, PG2 = 75 MW. Hence PG2 supplies load at bus3 (=300MW) and so Gen1 has to feed 225 MW load bus 3 and 200 MW load bus 2. This means 250 MW pass over line 1-2 (one circuit in operation) , 250-50= 50 MW pass over line 2-3, and 175 MW pass over line 1-3. The system is safe. Hence, the generation cost is equal to 425MW*MC1 + 75MW*MC2 = £ 13,000/h i.e. greater than that in a)

[5]

d) For the case of question b), determine the minimum Demand-Side Response (in busbar 2 or busbar 3) so that the power flow in the overloaded line are appropriately reduced in the case of circuit outages. Justify your answer. Calculate the corresponding generation operating costs.

[5]

For the case of question ii), we will determine the minimum Demand-Side Response (in busbar 2 or busbar 3) so that the power flow in the overloaded line is appropriately reduced in the case of circuit outages. Demand Response can be used as a measure of corrective control – changing demand leads to changes in power flows.

Demand Response from the loads at bus 2 and at bus 3 needs to be purchased so as to bring back the flows at line 1-2 within the limits, after a fault has occurred at line 1-2. The problem formulation becomes:

Minimize $DR_2 + DR_3$ subject to

$F_{12}^{(0)} + \frac{\Delta F_{12}}{\Delta P_2} DR_2 + \frac{\Delta F_{12}}{\Delta P_3} DR_3 \leq 250$ $i.e.$ $280 - 0.8DR_2 - 0.4DR_3 \leq 250$. Thus, given that 0.8 > 0.4, the demand response at bus 2 is more effective than that at bus 3 in reducing the flows at line 1-2. The solution is: $DR_2 = 37.5\ MW,\ DR_3 = 0\ MW$.

In this case, the load at bus 2 becomes 200 – 37.5 = 162.5MW, and the load at bus 3 is 300 MW. Hence, PG1 = 462.5 MW and PG2 =0 MW and the flow over line 1-2 is 250 MW (only one circuit is in operation) hence the flow over line 2-3 is 250 - 162.5 MW = 87.5 MW.

The flow over 1-3 is therefore 300 – 87.5 = 212.5 MW.

[5]

e) Discuss the differences between preventive control solution c) against corrective control solution d).

[3]

Demand Response is an alternative solution to generation redispatch. Preventive control of network flows through generation dispatch leads to higher generation costs (as rescheduling of generation following outage of a line cannot be delivered instantly). On the other hand, demand response can be delivered automatically, so generation dispatch is more efficient Thus demand response can be used as corrective control.

[3]

Question 2

Part A

a) List the four primary sources of big data in power systems.

[2]

The primary sources of big data in power systems are:
- Phasor Measurement Units (PMUs)
- Asset monitoring data from various system assets e.g. transformers
- Smart meters
- Electricity market data

b) Explain how asset health monitoring was carried out in the past and how it can benefit from big data.

[3]

Traditionally, asset maintenance has been scheduled on empirical rules (e.g. replace an asset every 10 years) and has been largely reactive (i.e. replace after a fault occurs). Big data analytics enable the real-time health monitoring of assets to improve efficiency. An example can be given for the case of monitoring of wind turbines to identify when maintenance should be carried out. By recording high-frequency observations of wind-speed and power output, a statistical model can be built to characterise the 'basecase' transfer function of a healthy asset. New observations can be compared against the basecase model and a 'distance metric' can be computed; divergence between measurements and model indicate need for maintenance.

c) Explain how dynamic stability assessment is carried out by Transmission System Operators (TSOs) currently, and how data from Phasor Measurement Units (PMUs) can be used to improve this.

[3]

Traditionally, security assessment has been carried out at a limited, ad-hoc basis, largely based on empirical rules. Big data analytics can be applied to data gathered from phasor measurement units (PMUs) to carry out large-scale security assessment and improve situational awareness. In particular, historical PMU data contain information on what operating points the operator should anticipate in the future. Simulations of faults can be carried out to identify whether each anticipated operating point is stable or unstable. Machine learning techniques are then used to produce security rules describing the system's security boundary.

Part B

a) Explain how governments aim to decarbonise electricity systems. Which are the resulting challenges for these systems and how can demand flexibility contribute to addressing these challenges?

[3]

Governments aim to decarbonise both generation and demand sectors of electricity systems.
Regarding the generation side, decarbonisation will be achieved through the wide integration of renewable and low-carbon generation sources, such as wind and solar generation. However, these generation sources are characterized by inherent variability and non-controllability. Given that demand is currently largely treated as an inflexible, uncontrollable load, the required flexibility for balancing the system and offering the required ancillary services is solely provided by conventional generators.

As the penetration of variable renewable generation increases, these conventional generators will be producing less energy, as absorption of the low-cost and CO2-free production of renewable generators will be prioritized in the merit order. However, they need to remain in the system and operate part-loaded as a back-up energy source and flexibility provider, since renewable generators not only have very limited capabilities to provide system balancing services, but they also make system balancing more challenging. This under-utilization of conventional generation assets implies that the cost efficiency of their operation will reduce. Furthermore, in cases where the flexibility of the conventional generation fleet is not sufficient, the last resort for system balancing lies in curtailing renewable generation. This implies that due to balancing challenges, renewable generation assets with high capital costs are also under-utilized and thus may not achieve their CO2 emissions reduction potential.
At the demand side, traditional technologies used in transport and heat sectors (internal combustion engines and gas / oil fired technologies respectively) are responsible for the emissions of a significant portion of the total greenhouse gases. In combination with the ongoing decarbonization of electricity generation, strong motives arise for the electrification of these sectors through the introduction of electric vehicles and electric heating respectively. However, due to the natural energy intensity and temporal demand patterns of transportation vehicles and heating loads, this paradigm change is accompanied by a significant increase of the total electricity demand and an even higher increase in the demand peaks, escalating the need for capital-intensive generation and network investments.
In this setting, flexible demand technologies, enabling temporal redistribution of electricity consumption patterns, can contribute in addressing these challenges by supporting system balancing, and therefore reducing system operating costs in a future with an increased penetration of renewable generation, as well as by limiting peak demand levels, and therefore avoiding capital-intensive investments in generation and network assets.

b) A simple power system, operating at a two-hour horizon, includes:
- a generator producing power s_t (MW) at hour t with a cost function $C_t(s_t) = 100s_t^2$ (£) and a maximum output limit $s^{max} = 10\text{MW}$
- inflexible demand appliances, consuming power $D_1 = 3.33\text{MW}$ (at hour 1) and $D_2 = 1.67\text{MW}$ (at hour 2)
- flexible demand consisting of 1000 identical appliances with continuously adjustable power d_t (kW), scheduling interval including hours 1 and 2, total energy required $E = 5\text{kWh}$ and maximum power limit $d^{max} = 6\text{kW}$

i) Assuming that price-based scheduling of the flexible demand appliances is employed, with the prices at hour 1 and 2 being $\lambda_1 = 2£/kWh$ and $\lambda_2 = 1£/kWh$, what is the resulting power demand of each of the flexible demand appliances at hours 1 and 2? What is the total generation cost? [3]

Each flexible demand appliance aims at minimizing its electricity cost, which is expressed by the following function:
$C^{FD} = \lambda_1 * d_1 + \lambda_2 * d_2$ (£)
Therefore, it will obtain as much energy as possible at the hour with the lowest price and the rest at the hour with the highest price:
$d_2 = \min(E, d^{max}) = 5\text{kW}$ and $d_1 = E - d_2 = 0\text{kW}$
The system operator schedules the generator to meet the resulting total demand:
$s_1 = D_1 + 1000 * d_1 = 3.33\text{MW}$ and $s_2 = D_2 + 1000 * d_2 = 6.67\text{MW}$

The total generation cost is $100(D_1 + 1000 * d_1)^2 + 100(D_2 + 1000 * d_2)^2 = £5558$

ii) Assuming that, apart from the prices λ_1 and λ_2, a flexibility penalty α (measured in $£/kWh^2$) is applied to the square of each flexible appliance's power demand, determine the resulting power demand of each of the flexible demand appliances at hours 1 and 2 as a function of α.

[3]

The electricity cost function of each flexible demand appliance now changes to:

$C^{FD} = \lambda_1 * d_1 + \lambda_2 * d_2 + \alpha * {d_1}^2 + \alpha * {d_2}^2$ (£)

Although this function includes two decision variables d_1 and d_2, we can express their relationship through their energy requirement constraint:

$$d_2 = E - d_1$$

By substituting the latter equation in the above cost function, we get:

$C^{FD} = \lambda_1 * d_1 + \lambda_2 * (E - d_1) + \alpha * {d_1}^2 + \alpha * (E - d_1)^2$ (£)

In order to determine the power demand of each flexible demand appliance, we need to set the derivative of this cost function to be equal to zero:

$$\frac{dC^{FD}}{dd_1} = 0 \rightarrow \lambda_1 - \lambda_2 + 2 * \alpha * d_1 + 2 * \alpha * d_1 - 2 * \alpha * E = 0$$

From which we obtain:

$$d_1 = \frac{2 * \alpha * E - \lambda_1 + \lambda_2}{4 * \alpha} \rightarrow d_1 = \frac{10 * \alpha - 1}{4 * \alpha}$$

Consequently, we also obtain:

$$d_2 = E - d_1 \rightarrow d_2 = 5 - \frac{10 * \alpha - 1}{4 * \alpha}$$

iii) Varying the flexibility penalty α in the range $0.1£/kWh^2$ to $0.5£/kWh^2$ with a step of $0.1£/kWh^2$, calculate its most suitable value from the system perspective, by employing a relevant matrix and explaining your calculations properly.

[3]

In order to calculate the most suitable value of the flexibility penalty α from the system perspective (i.e. minimising the generation cost in the system), the requested candidate values of α are tried out as demonstrated in Table 1 below. For each value of α, the following quantities are calculated:

- Demand of each flexible demand appliance at each of the two hours, based on the functions determined in question b.ii above
- Total demand (and total production) in the system at each of the two hours: $s_1 = D_1 + 1000 * d_1$ and $s_2 = D_2 + 1000 * d_2$
- Total generation cost: $100(D_1 + 1000 * d_1)^2 + 100(D_2 + 1000 * d_2)^2$

Table 1: Scheduling of flexible demand appliances with different candidate values of α

α	d_1 (kW)	d_2 (kW)	s_1 (MW)	s_2 (MW)	Gen. cost (£)
0.1	0	5	3.33	6.67	5558
0.2	1.25	3.75	4.58	5.42	5035
0.3	**1.667**	**3.333**	**5**	**5**	**5000**
0.4	1.875	3.125	5.205	4.795	5008
0.5	2	3	5.33	4.67	5022

The most suitable value of the flexibility penalty α minimising the generation cost in the system is $\alpha = 0.3$

Question 3

a) Discuss how you might quantitatively characterise past and future performance for the distribution and transmission network. Provide examples of performance measures for the distribution and the transmission networks and discuss why they are different.

[3]

The degree of performance of the distribution and transmission network is characterised by performance metrics describing the reliability of the system. A distinction is made between past and future performance:

- Past has happened (retrospective): performance metrics are well specified
- Future has not happened yet (prospective): performance metrics are random variables.

The future performances are characterised by risk indices of the random performance metrics and risk profiles.

Examples of performance measures for the distribution and transmission network are:

- Transmission
  - LOLP/ LOLE
  - EENS
  - LOLF
- Distribution
  - CML / SAIDI
  - CI / SAIFI
  - ENS

MARKING:

– 2 point: 1 point to correctly define performance metrics, 1 additional point for a more comprehensive discussion on the difference between past and future performance.

– 1 point: correctly providing examples of performance measures for the distribution and transmission network.

b) Consider a section of a distribution network serving 9500 customers. The outage events occurred in one year are reported in Table 3.1, except for an outage that affected 2000 customers scattered in a large area. In this last outage lasting for 1 hour the customers are not reconnected all at the same time. Assume that their reconnection on average followed a linear trend as a function of time. Compute CML and CI for this section of the network. Discuss the difference between CML and SAIDI.

Customers Affected	**Duration**
1200	30 minutes
300	5 hours
2500	2 minutes
900	50 minutes
20	36 hours
700	10 minutes

Table 3.2- Outage events

[3]

Due to the 3-minute minimum requirement for CML and CI, the recorded event that affected 2500 customers for 2 minutes will be excluded from CML/CI calculations.

From the outage affecting 2000 customers for 60 minutes we need to exclude the customers reconnected in less the 3 minutes accordingly the linear trend. The number of affected customers $C_m(t)$ satisfies the following trend as function of the time

$$C_m(t) = -\frac{2000}{60}t + 2000$$

The number of customers disconnected after 3 minutes is 1900. Consequently, the contribution of this outage to CML is given by area under the curve $y(t) = min(1900, C_m(t))$ between $t = 0$ and $t = 60$ minutes that is $1900 * 3 + 1900 * \frac{57}{2} = 59850$.

Then the CML is

$$\text{CML} = \frac{\sum_{events} N_{affected} T_{minutes}}{N_{customers}}$$

$$= \frac{1200 * 30 + 300 * 5 * 60 + 900 * 50 + 20 * 36 * 60 + 700 * 10 + 59850}{9500}$$

$$= 29.58 \text{ minutes}$$

$$\text{CI} = 100 \times \frac{\sum_{events} N_{affected\ customers}}{N_{customers}}$$

$$= 100 \times \frac{3500 + 700 + 5000 + 1100 + 10 + 900}{12000} = 52.84$$

SAIDI, similarly to CML, is defined as the average customer minutes lost per customer per year. However, SAIDI, differently from CML in which only interruptions lasting three minutes or longer are considered, must include all the events.

MARKING:

- 1 point: correctly excluding the 2-minute event
- 1 point: providing correct expressions for CML and CI and correct distinction between CML and SAIDI
- 1 point: correct numbers

c) Consider the distribution network in Figure 3.1

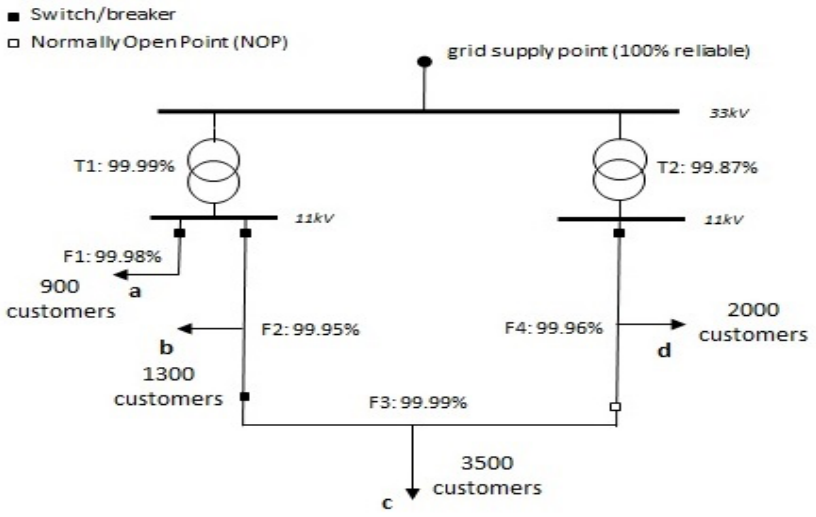


Figure 3.1- Network diagram

The number of customers at each load point (a, b, c, d) is reported in Figure 3.1. The availabilities of the transformers (T1, T2) and the feeders (F1, F2, F3, F4) are reported in Figure 3.1. Assume that all components failures are independent and they all last

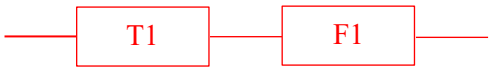
more than three minutes. The individual capacity of T1 and T2 are always sufficient to serve all customers at load points a, b c and d. The network is radially operated but the normally open points (NOPs) can ensure some redundancy. They can be closed to resupply load points in case of faults in the network.

i) Construct reliability block diagrams for the power supply to each individual load point and calculate the unavailability of power in each case. Assume all interruptions are longer than 3 minutes. Determine the CML for this network.

[5]

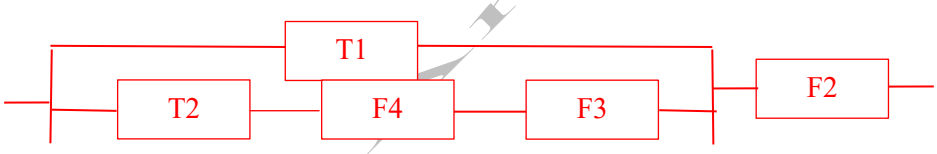
Let p(x) denote the availability of component x and a_x the probability that customers at load point x are connected to a grid supply point.
The reliability block diagrams to each individual load point and associated unavailability q_x in each case are as follows

– Load a



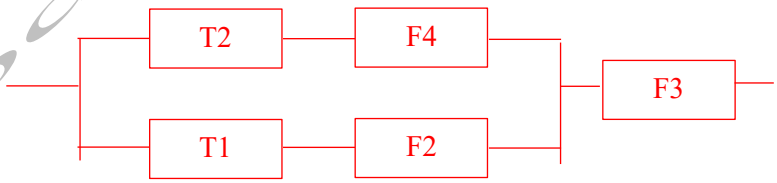
$$q_a = 1 - a_a = 1 - p(T1)p(F1) = 1 - 0.9997 = 0.0003$$

– Load b



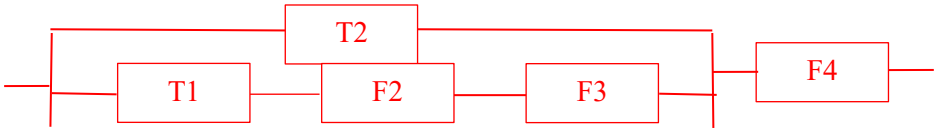
$$q_b = 1 - a_b = 1 - p(F2)(1 - (1 - p(T1))(1 - p(T2)p(F3)p(F4))$$
$$= 1 - 0.9995 = 0.0005$$

– Load c



$$q_c = 1 - a_c = 1 - p(F3)(1 - (1 - p(T2)p(F4))(1 - p(T1)p(F2)))$$
$$= 1 - 0.9999 = 0.0001$$

– Load d



$$q_d = 1 - a_d = 1 - p(F4)(1 - (1 - p(T2))(1 - p(T1)p(F3)p(F2)))$$
$$= 1 - 0.9996 = 0.0004$$

Model answers

$$CML = 8760h \cdot \frac{60min}{h} \cdot \frac{\sum_{loadpoints} N_{customers\ at\ loadpoint} \cdot unavailability_{loadpoint}}{N_{customers}}$$

$$= 8760h \cdot \frac{60min}{h} \cdot \frac{900 \cdot q_a + 1300 \cdot q_b + 3500 \cdot q_c + 2000 \cdot q_d}{7700} = 144.16\ min$$

MARKING:

- 4 points: correct expressions for a_a, a_b, a_c and a_d (1 point for each load point)
- 1 points: correct expression and evaluation of CML

ii) Assume that an interruption incentive scheme is in place and there is an adjustment on the revenues of the distribution company, based on the actual CML index and its target CMLTARGET=135 minutes. Such adjustment ΔR can be expressed as:

$$\Delta R = \text{k} * (CMLTARGET - CML) \quad (£)$$

(constant k is assumed to be positive).

The company has the possibility of introducing switches S1 and S2 to isolate the transformers T1 and T2 from the bus B1 and B2 at 11k and place a new feeder connecting B1 to B2 with reliability $\rho = 99.99\%$ as illustrated in Figure 3.2. Assume that the reliability of the two new switches S1, S2 and the buses B1 and B2 can be approximated as 1. What would be the maximum cost of this upgrade that would justify the investment?

[6]

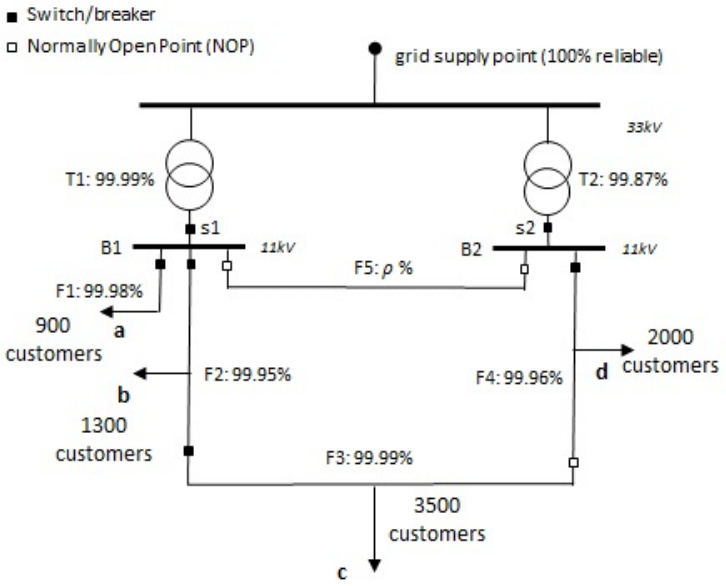
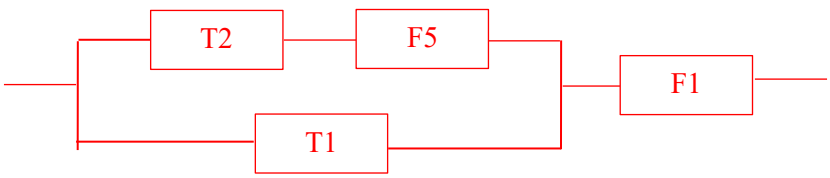


Figure 3.2: Modified Network

All calculations at the previous points are repeated assuming that S1, S2, $B1$ and $B2$ have full reliability (p(S1)=p(S2)=p(B1)= p(B2)=1).

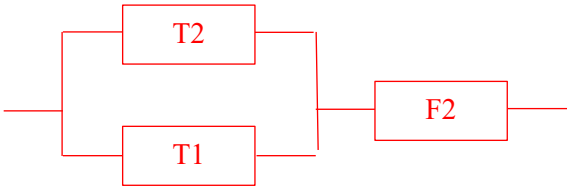
Load a



$$a_a = \left(1 - \left(1 - p(T1)\right)\left(1 - p(T2)p(F5)\right)\right) p(F1) = 0.9998$$

Load b

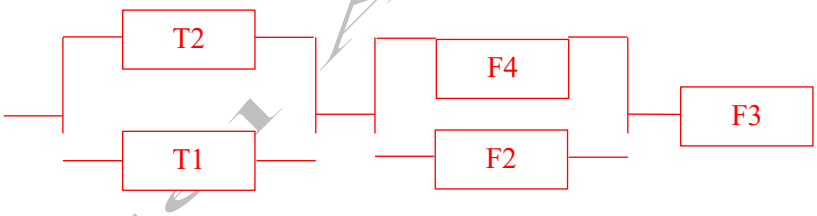
The probability of success for load b can be computed combining the probability of success when the feeder F5 is not working and the probability of success when F5 is working. If F5 is working the RBD for load b is as follows



$$a_b = (1 - p(F5))p(F2)\left(1 - \left(1 - p(T1)\right)\left(1 - p(T2)p(F3)p(F4)\right)\right) + p(F5)p(F2)\left(1 - \left(1 - p(T1)\right)\left(1 - p(T2)\right)\right) = 0.9995$$

Load c

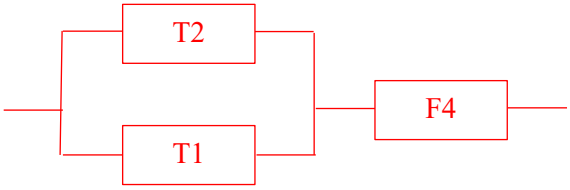
The probability of successfully serve load c can be computed combining the probability of success when the feeder F5 is not working and the probability of success when F5 is working. If F5 is working the RBD for load c is as follows



$$a_c = \left(1 - p(F5)\right)p(F3)\left(1 - \left(1 - p(T2)p(F4)\right)\left(1 - p(T1)p(F2)\right)\right) + p(F5)p(F3)(1 - \left(1 - p(F2)\right)\left(1 - p(F4)\right))(1 - \left(1 - p(T1)\right)\left(1 - p(T2)\right)) = 0.9999$$

Load d

The probability of successfully serve load d can be computed combining the probability of success when the feeder F5 is not working and the probability of success when F5 is working. If F5 is working the RBD for load c is as follows



$$a_d = (1 - p(F5))p(F4)\left(1 - (1 - p(T2))(1 - p(T1)p(F3)p(F2))\right)$$
$$+ p(F5)p(F4)\left(1 - (1 - p(T1))(1 - p(T2))\right) = 0.9996$$

The new expected value of CML is:

$$E[CML] = 135.80\ minutes$$

The maximum justifiable investment cost $Cmax$ can be expressed:

$$Cmax = \mathrm{k} * (144.16 - 135.80)\ £$$

MARKING:

- 4 points: correct expressions for a_a, a_b, a_c and a_d (1 point for each load point)
- 1 points: correctly determining the required procedure
- 1 point: correct numbers

d) Consider a nonrepairable parallel structure of two independent components with lifetimes T_1 and T_2 independent and exponentially distributed with constant failure rates λ_1 and λ_2. Compute the mean time to failure of the parallel structure. Is the failure time T of the parallel structure exponentially distributed?
(Suggestion*: use the survival function* $S_T(t)$ *and note that* $[tS_T(t)]_0^\infty = 0$)

[3]

The MTTF is, by definition, the mean value of the stochastic failure time T

$$MTTF = E[T] = \int_0^\infty t\, f_T(t)dt$$

where $f_T(t)$ is the density function of the parallel structure. Since $f_T(t) = -\dot{S}_T(t)$ the MTTF can be expressed as follows

$$E[T] = -\int_0^\infty t\dot{S}(t)dt = [-tS_T(t)]_0^\infty + \int_0^\infty S_T(t)dt = \int_0^\infty S_T(t)dt$$

The survival function of the parallel structure of two independent component is

$$S_T(t) = 1 - \left(1 - e^{-\lambda_1 t}\right)\left(1 - e^{-\lambda_2 t}\right)$$

since the probability of the component i of being available at time t is $e^{-\lambda_i t}$ (exponential distribution).

Consequently, the MTTF of the parallel structure is

$$MTTF = \int_0^\infty S_T(t)dt = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2}$$

The simplified expression of survival function $S_T(t)$ of the parallel structure is as follows

$$S_T(t) = e^{-\lambda_1 t} + e^{-\lambda_2 t} - e^{-(\lambda_1 + \lambda_2)t}$$

Note that $S_T(t)$ cannot be recast as $e^{-\lambda t}$ for some positive constant λ, and consequently the stochastic failure time T is not exponentially distributed.

MARKING:
- 1 point: correctly giving the definition and expression of MTTF
- 1 point: correctly computing the MTTF
- 1 point: correctly saying that the distribution of the parallel structure is not exponential

Model Answers

Question 4

a) Monte Carlo simulation is used to estimate the expected amount of interruptions.

i) The application of Monte Carlo Simulations relies on three fundamental theories. Please state these three theories and how they support the application.

[3]

1) The (Strong) Law of Large Numbers guarantees that $\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \mathrm{Im}(s_i) = E[\mathrm{Im}]$, which is used to estimate the true risk.

[1]

2) The central limit theorem tells us that, for large n: $\overline{\mathrm{Im}} \xrightarrow{d} \mathcal{N}\left(\mu_{Im}, \frac{\sigma^2_{Im}}{n}\right)$, which is used to estimate the MC error distribution.

[1]

3) Sample standard deviation is used to estimate the true standard deviation $\hat{\sigma}_{Im} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\mathrm{Im}(s_i) - \frac{1}{n}\sum_{j=1}^{n}\mathrm{Im}(s_j)\right)^2}$

[1]

ii) After N=10,000 samples the best estimate of the interruption is 108,762 MW, with a standard error of 2581 MW. How many samples will be required (approximately) to reduce the coefficient of variation to 0.3%?

[2]

The coefficient of variation is defined as $\mathrm{CoV} = \mathrm{se}/\hat{r}$. The requirement is achieved when the standard error (σ') satisfies
σ'/108762=0.3%

[1]

Hence, σ'=326 and the required number of samples becomes
N'=N*(2581/326)^2=626817

[1]

iii) What are the advantages and disadvantages of Monte Carlo simulation?

[2]

Advantages

- MC simulations are often easy to implement – even for complex systems. Therefore, they provide a **rapid** and **intuitive** way to 'play' with uncertainties and see how they affect the system.
- The performance of MC simulations is **unaffected by the dimensionality of the system**. Systems with many sources of uncertainty are analysed with the same ease as simpler systems.

[1]

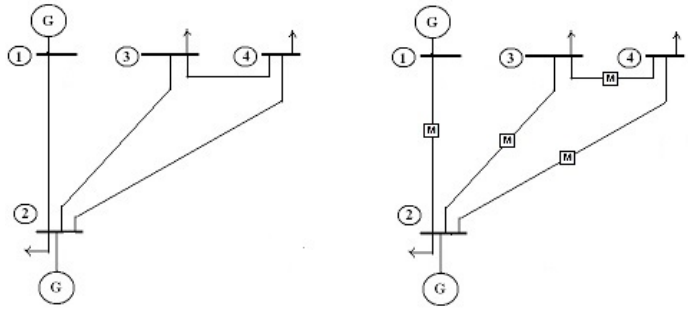
Disadvantages

- It is a **stochastic method**, so the results change every time you run a simulation.

- Its speed of convergence is relatively slow. Using MC simulations, you quickly get a reasonable answer, but getting a good answer can take a long time.

[1]

b) Consider the 4-bus system consisting of load and generators connected as below. The system is measured by meters placed on each branch. Assume PU reference is 1MW.



PER UNIT REACTANCES

X12	0.1
X23	0.2
X24	0.35
X34	0.5

PER UNIT POWER FLOWS

Z12	-70
Z23	37.6
Z24	32.4
Z34	7.6

Figure 4.1 4-bus system with branch meter placement and unit reactance

i) Construct the branch incidence matrix **H** for this system. To do this you will consider the incidence between each bus. Hint: Can be done either by direct inspection of the branches or by using a combination of the incidence **I** and branch diagonal **D** via **H=D∗I**.

[1]

$$H = \begin{bmatrix} 10 & -10 & 0 & 0 \\ 0 & -5 & 5 & 0 \\ 0 & -2.87 & 0 & 2.87 \\ 0 & 0 & -2 & 2 \end{bmatrix}$$

[1]

ii) An attacker would like to convince the system operator that line Z23 is overloaded. The power flow limit for this line is 50MW. Assume bus 1 is the reference bus (θ_1=0), the estimated state for the original measurement are $\theta_2 = -7.00$, $\theta_3 = 0.524$, $\theta_4 = 4.333$. A naive attacking strategy is to change the measure of Z_{23} to 50 MW, which results in a new estimated states $\theta_2 = -7.00$, $\theta_3 = 2.86$, $\theta_4 = 5.8$. Please calculate the residuals for both the original measurements and the attacked measurements and comment on whether this attack strategy will be successful if the alarm on residual is set at 0.1.

[4]

The residual is the difference between the measured values defined by measurements and the estimated values:

$$r = Zm - Z_est = z - Hx$$

Measured values are as given;

$$Zm = [-70,37.62,32.4,7.62]$$

Which using the given values looks like
Z_est = [10*-(7+0), 5*(0.524+7) ,2.85*(4.33+7),2*(4.33-0.524)]
= [-70,37.62,32.38,7.62]

Residual is the difference between the calculated and measured.
r = sum (absolute(Z_m – Z_est)) = 0 (or close to)

[2]

This same calculation is repeated for the new vector with a resultant residual value of 3.6 (significantly larger than the alarm vector value specified in the question) therefore we would expect the attacking strategy to fail.

[2]

iii) If the attack in ii) is not likely to be successful, how can the attacker design the attack to drive the overload without being detected? What is the minimum number of measurements the attacker needs to change in order to achieve this aim? Where appropriate, please use matrix formulations to justify your answer.

[5]

Avoiding Detection

The attacker will need to consider the system topology matrix H in order to attack without being detected.
The residual is a function of the measured flows and estimated values such that:

$$r = z_m - Hx$$

x values are calculated using the WLS estimator which is based on the system topology.
Therefore, we can structure our attack vector based on the system topology such that

$$z_a = z + Hc$$

Where c is a vector of length n indicating the bias we wish to add to the system. This will give a residual value of

$$r_a = z + a - H(x+c)$$

And as our z was structured using the WLS estimator (incorporating H) the a component and c will cancel.

[2]

Best Target

The attacker can achieve his aim either by changing state angle at bus 2 or bus 3 via altering meter measurements. As bus 3 has less interconnections the number of meters required to remain stealthy is only 2 compared with 3 for bus 2. The student can either spot this by inspection or by referring to the column matrix of H which is sparser for column 3 vs column 2.

[3]

c) Explain the Willingness-to-Pay (WTP) and Willingness-to-Accept (WTA) approaches to evaluating customer interruption costs. Do you expect similar values when using WTA and WTP? Explain which approach you consider most suitable as a basis for interruption costs models.

[3]

1) "Willingness to Pay" (WTP): Ask respondents how much they would be willing to pay to avoid a given interruption scenario. "Willingness to Accept" (WTA): Ask respondents the level of payment they would require accepting an interruption scenario.

[1]

3) To a rational decision maker, WTA and WTP would be identical, assuming the costs are not a significant part of the available budget. However, for human subjects WTA is usually much larger than WTP, due to the endowment effect. This is due to the asymmetric valuation of gains and losses. Any consistently argued preference for WTA or WTP, or discussion about their relative strengths/weaknesses is acceptable.

[2]

Model Answer